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Research assignment 1

Data analytics

- ① A database is an organised collection of data that can be stored, accessed and managed easily. Some examples to consider include customer records, student information and bank statements from accounts.
- ② A data warehouse stores large amounts of historical data from various sources. Unlike normal database, it is mainly used for analysis & reporting.
- ③ A smaller part of a data warehouse focused on a specific department/section such as Sales or finance; mainly used by departments.
- ④ A data lake stores large amounts of raw data in different formats. Businesses prefer this when they need flexibility & want to analyse data later.
- ⑤ Structured data :- Organised in rows & columns, example :- Spreadsheets

Semi structured data :- has some organisation but not fixed tables.

example :- JSON

Unstructured data :- has no fixed structure
Example :- Video, images

⑥ Data modelling is planning how data will be organised and related. It helps create efficient Systems

- * Two types of data modelling
 - Conceptual
 - logical

⑦ Data mining is finding useful patterns & relationships in large datasets. Examples include

- Detecting credit card fraud
- Identifying products customers often buy together

⑧ A DBMS is a Software ~~are~~ used to create, store and manage databases. Examples :-

- * MySQL :- Websites & App's
- * PostgreSQL :- Complex App's
- * Oracle :- Large businesses
- * MongoDB :- Applications using flexible document data

⑨ ETL means Extract; Transform; Load. Data is collected from sources, ~~are~~ cleaned/changed into a useful format, and then loaded into a database or Warehouse. It prepares data for analysis.

⑩ Common Data Storage formats

- * CSV :- Simple table format, good for Spreadsheets
- * JSON :- Stores structured data, commonly used with API's
- * Parquet :- efficient for large datasets & analysis
- * Avro :- Useful for storing & transferring large amounts of structured data.